Course Name: Numerical Methods

Course Code: **Internal Marks: 25** **External Marks: 75** **Credit: 4**

UNIT-I

Roots of Equations:- Bisections Method, False Position Method, Newton's Raphson Method, Rate of convergence of Newton's method.

UNIT-II

Interpolation and Extrapolation:- Finite Differences, The operator E, Newton's Forward and Backward Differences, Newton's dividend differences formulae, Lagrange's Interpolation formula for unequal Intervals, Gauss's Interpolation formula, Starling formula, Bessel's formula, Laplace-Everett formula.

UNIT-III

Numerical Differentiation Numerical Integration:- Introduction, direct methods, maxima and minima of a tabulated function, General Quadratic formula, Trapezoidal rule, Simpson's One third rule, Simpson's three- eight rule.

UNIT-IV

Solution of Linear Equation:- Gauss's Elimination method and Gauss's Siedel iterative method.

UNIT-V

Solution of Differential Equations:- Euler's method, Picard's method, Fourth-order Ranga – Kutta method.